Bayesian Statistical Analysis for Charging Policy Influence on Expected Battery Life Cycle

Hierarchical Negative Binomial Regression

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Introduction

Lithium-ion battery cycle life is a critical constraint for energy storage solutions at both grid and small-scale device levels. The number of cycles a battery achieves depends on multiple factors: operating conditions, thermal management, charging policy, and manufacturing variability, making precise prediction inherently challenging. Uncertainty in these operating conditions and cell-to-cell heterogeneity render deterministic models unreliable for capturing the true range of outcomes. Yet energy companies and organizations that deploy or manufacture batteries rely on accurate cycle-life estimates to ensure grid stability and operational reliability. Thus, developing robust methods to quantify expected cycle life is critical for informed decision-making across the energy storage sector.

The aim of this analysis is to build a statistical model for the influence of charging policy on the number of expected life cycles. Then use this model to predict the influence of the three different charging policy on the expected number of life cycles. The three policies are

Table 1: Summary of different charging policies

Charging Policy	Charging rate	charging cut-off	discharging rate
Conservative	3.6 c	40%	3.6 c
Balanced	3.6 c	60%	3.6 c
Aggressive	7.5 c	80%	5.5 c

We use the publicly available Severson et al. LFP/graphite battery dataset. Where 124 cells have been charged and discharged following different policies. The rest of this report is structured as following: we first start by exploring the data set, cleaning and reformatting the entries into useful form for the analysis. Then we explain the modeling strategy we followed. After conducting model convergence analysis, we compare the different models used. In the final section, we use the model to answer the question: What is the probability of attaining a certain number of life cycles under the three different charging policies indicated in table 1.

Contents

1 Data Exploration

Table 2 shows the first 5 lines of the data. The data is grouped by **Batch Data**, which suggest the presence of different batches. Furthermore, the data was used into training a machine learning algorithm hence column two **Dataset**. **Cycle life** is our observables and **Charging policy** contains the explanatory variables. In the final column the explanatory variables need to be separated for the purpose of our analysis.

Table 2: Example of data entries

Cell barcode	**Dataset**	**Batch date**	**Cycle life**	**Charging policy**
EL150800460514	Prim. Test	5/12/17	1852	3.6C(80%)-3.6C
EL150800460486	Train	5/12/17	2160	3.6C(80%)-3.6C
EL150800460623	Prim. Test	5/12/17	2237	3.6C(80%)-3.6C
EL150800464977	Train	5/12/17	1434	4C(80%)-4C
EL150800464865	Prim. Test	5/12/17	1709	4C(80%)-4C

There is three batches corresponding to three different dates. Figure 1 is a box plot showing the

date	batch	number of cells
4/12/18	1	40
5/12/17	2	41
6/30/17	3	43

Table 3: The three different data batches

variability of the number of life cycles between batches. Batches 1 and 2 have a close median to 1000 while batch three has median close to 500, this suggests that there is strong variability between batches which needs to be taken into account during modeling. Figure 2 summaries the covariance of the

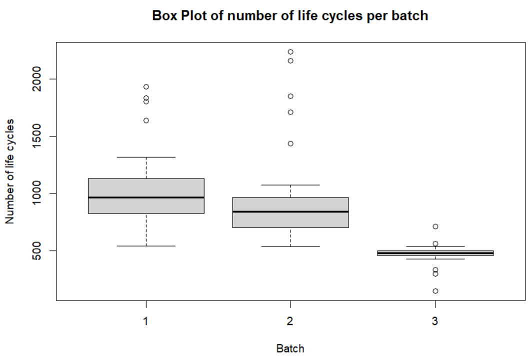
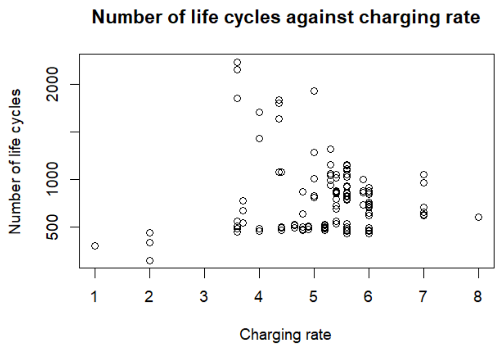
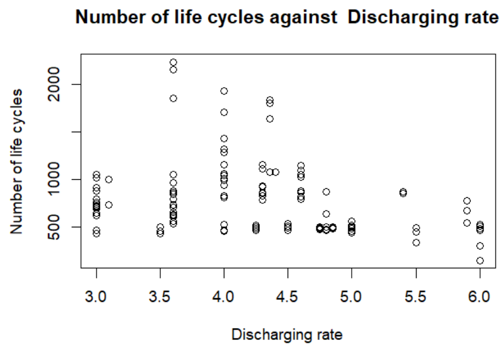


Figure 1: Life cycle per batch

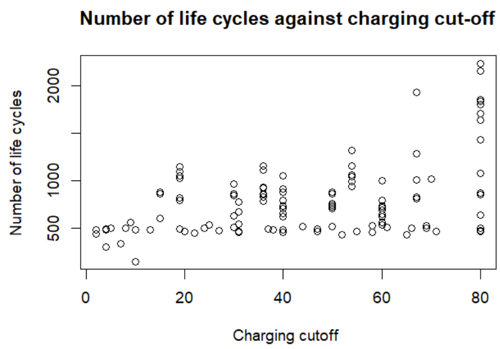
number of cycles of life with the three different covariates. Figure 2a seems to suggest that a moderate charging rate is favorable, while figure 2b implies that a smaller discharging rate increases the life of the battery. Finally increase the the charging cut-off seems to increase the life of the battery as figure 2c shows.



(a) Charging Rate



(b) Discharge Rate



(c) Charging Cut-off

Figure 2: Explanatory parameters influence on number of cycles of life

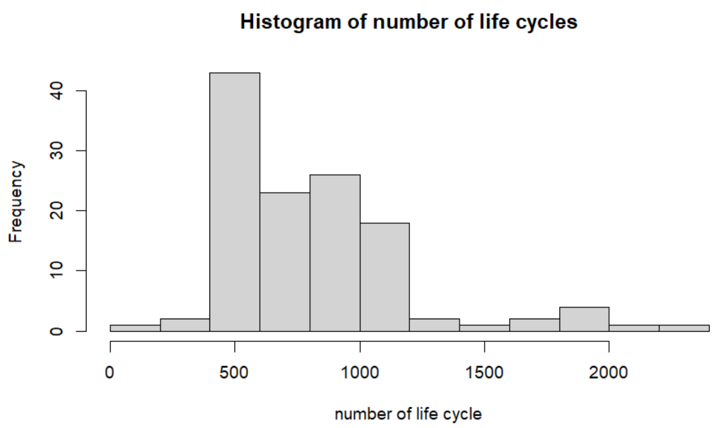


Figure 3: Histogram of number of life cycles

Table 4: Overdispersion across batches

batch	mean	variance	fraction
1	1032	95232.97	92.28
2	921	160613.3	174.32
3	473	5899.55	12.46

The histogram 3 shows that the number of life cycles is centered between 500 and 1000. The number of life cycles is a count, thus it seems reasonable to suggest a Poison regression. However, number of life cycles is inherently overdispered as shown is table 4.

Because in poison regression the mean is equal to the variance, it would not account for this characteristic of the data. Thus we will chose the negative binomial distribution, where we can control the mean and the variance separately.

2 Model Postulation

Following the conclusion of the previous section. We chose to model the number of cycle of life as a negative binomial. Furthermore, the differences between the batches justifies the need for a hierarchical model.

$$
\begin{aligned}
y_i \mid b_i, \boldsymbol{\beta}, \boldsymbol{x}, r &\sim \text{NegBin}(\phi_i, r) \\
\log(\mu_i) &= \alpha_{b_i} + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 \\
\phi_i &= \frac{r}{\mu_i + r} \\
\beta_j &\sim \mathcal{N}(\mu_\beta, \tau_\beta) \quad \text{for } j = 0, 1, 2, 3 \\
r &\sim \text{Gam}(a, b) \\
\alpha_{b_i} &\sim \mathcal{N}(\mu_\alpha, \sigma_\alpha) \text{for} b_i = 1, 2, 3 \\
\mu_\alpha &\sim \mathcal{N}(h_\mu, \sigma_\mu) \\
\sigma_\alpha &\sim \textbf{IGam}(k_1, k_2)
\end{aligned}
$$

Table 6 summaries the different parameters present in the model and there values. We use a linear regression model on the mean of the negative binomial with a random intercept for each batch. Because the mean needs to be positive we use a **log link function**. Furthermore we chose to model the mean instead of the probability ϕ_i, to ensure easy interpretation of the parameters of the model. Finally, we will compare three different models: models two and three include an interaction term between the charge and discharge, while model one does not. Models 1 and two have relatively non informative priors on the β coefficient, while model 2 is more centered around zero. We will perform a model sensitivity analysis using models 2 and 3.

Table 5: Models definition

	model 1	model 2	model 3
interaction between charge and discharge	no	yes	yes
Prior	not skeptical	skeptical	not skeptical

Table 6: Bayesian Hierarchical Model: Variables and Specifications

Notation	**Description**	**Distribution/Value**
	Likelihood	
y_i	Cycle life for cell i	Observed data
b_i	Batch indicator for cell i	$\{1, 2, 3\}$
x_1	Charging cut-off (% DoD)	Observed predictor
x_2	Charging rate (C-rate)	Observed predictor
x_3	Discharging rate (C-rate)	Observed predictor
x_4	Interaction parameter: Charge * discharge	Observed predictor
	Link Function & Probability	
μ_i	Expected cycle life	$\log(\mu_i) = \alpha_{b_i} + \sum_{j=0}^{3} \beta_j x_j$
ϕ_i	NegBin probability parameter	$\frac{r}{\mu_i + r}$
r	NegBin dispersion parameter	$\sim \text{Gam}(0.5, 0.5)$
	Regression Coefficients (Level 1)	
β_1	Coefficient for charging cut-off	$\sim \mathcal{N}(\mu_\beta, \tau_\beta)$
β_2	Coefficient for charging rate	$\sim \mathcal{N}(\mu_\beta, \tau_\beta)$
β_3	Coefficient for discharging rate	$\sim \mathcal{N}(\mu_\beta, \tau_\beta)$
β_4	Interaction coefficient	$\sim \mathcal{N}(\mu_\beta, \tau_\beta)$
	Batch Random Effects (Level 2)	
α_{b_i}	Random intercept for batch b_i	$\sim \mathcal{N}(\mu_\alpha, \sigma_\alpha)$ for $b_i \in \{1, 2, 3\}$
μ_α	Prior mean of batch intercepts	$\sim \mathcal{N}(h_\mu, \sigma_\mu)$
σ_α	Prior std. dev. of batch intercepts	$\sim \text{IGam}(k_1, k_2)$
	Hyperparameters (Level 3)	
μ_β, τ_β	Coefficient prior mean & precision	$\mu_\beta = 0, \tau_\beta = 1/100\ (1/10.0)$
h_μ, σ_μ	Prior for batch intercept mean	$h_\mu = 6.0, \sigma_\mu^2 = 100$
k_1, k_2	Inverse Gamma shape/scale	$k_1 = 0.5, k_2 = 0.5$
a, b	Gamma shape/rate for r	$a = 0.5, b = 0.5$

3 Model fitting and checking

Before fitting the models. We first divide the data into two parts: the first parts is used to fit the models, while the second part is used to see how well the models generalize to other data sets. [1]

3.1 Convergence analysis

We run the MCMC for three different chains for each model with a burn-in equal to $1e4$. Then we iterate the chain for $1e5$.

Table 7: Model 1: Gelman-Rubin Convergence Diagnostic and Effective Sample Size

Parameter	$\hat{R}$ (Point)	$\hat{R}$ (Upper C.I.)	**ESS**
β_1 (Charging cut-off)	1.03	1.10	514.4
β_2 (Charging rate)	1.05	1.17	188.6
β_3 (Discharging rate)	1.04	1.13	110.0
α_1 (Batch 1)	1.06	1.19	48.3
α_2 (Batch 2)	1.06	1.20	55.0
α_3 (Batch 3)	1.06	1.19	46.2
μ_α (Intercept prior mean)	1.02	1.07	154.1
r (Dispersion)	1.00	1.00	13168.3
σ_α (Batch std. dev.)	1.00	1.01	14498.9
Multivariate PSRF	1.04		

Table 8: Model 1: Posterior Summary Statistics

Parameter	**Mean**	**SD**	**2.5%**	**Median**	**97.5%**
β_1	0.389	0.171	0.061	0.386	0.730
β_2	-0.112	0.046	-0.202	-0.112	-0.021
β_3	-0.173	0.072	-0.301	-0.175	-0.029
α_1	8.085	0.536	7.009	8.083	9.027
α_2	7.867	0.513	6.845	7.865	8.781
α_3	7.370	0.526	6.317	7.370	8.295
μ_α	7.762	0.788	6.197	7.765	9.237
r	12.514	1.999	8.919	12.416	16.714
σ_α	0.886	0.573	0.367	0.732	2.355

All Gelman-Rubin $\hat{R}$ statistics are below 1.10, indicating acceptable convergence for both models. However, **Model 3 demonstrates superior convergence properties**:

- **Multivariate PSRF:** Model 3 achieves 1.01 (excellent) versus Model 1's 1.04 (acceptable)
- **Batch effects:** ESS for α_j improves substantially in Model 3 (106–121 vs. 46–55), indicating reduced autocorrelation
- **Dispersion parameter:** ESS for r increases 5-fold in Model 3 (60,852 vs. 13,168)
- **Posterior variance:** σ_α achieves ESS = 132,073 in Model 3, allowing precise uncertainty quantification

[1] code is available in https://github.com/Amine-Physics/Bayesian-analysis-of-battery-life-cycle

Table 9: Model 3: Gelman-Rubin Convergence Diagnostic and Effective Sample Size

Parameter	$\hat{R}$ (Point)	$\hat{R}$ (Upper C.I.)	**ESS**
β_1 (Charging cut-off)	1.00	1.01	2974.9
β_2 (Charging rate)	1.01	1.03	79.5
β_3 (Discharging rate)	1.01	1.03	80.5
β_4 (Additional covariate)	1.01	1.02	163.9
α_1 (Batch 1)	1.01	1.04	121.7
α_2 (Batch 2)	1.01	1.03	106.1
α_3 (Batch 3)	1.01	1.04	112.3
μ_α (Intercept prior mean)	1.01	1.02	214.2
r (Dispersion)	1.00	1.00	60852.6
σ_α (Batch std. dev.)	1.00	1.00	132073.2
Multivariate PSRF	1.01		

Table 10: Model 3: Posterior Summary Statistics

Parameter	**Mean**	**SD**	**2.5%**	**Median**	**97.5%**
β_1	0.189	0.156	-0.122	0.190	0.492
β_2	-0.861	0.152	-1.199	-0.847	-0.578
β_3	-1.029	0.177	-1.419	-1.015	-0.698
β_4	0.171	0.033	0.109	0.168	0.241
α_1	11.961	0.903	10.308	11.890	13.950
α_2	11.838	0.911	10.174	11.762	13.856
α_3	11.286	0.904	9.626	11.214	13.280
μ_α	11.673	1.098	9.638	11.630	13.961
r	16.626	2.700	11.775	16.475	22.328
σ_α	0.895	0.654	0.367	0.730	2.400

Table 11: Convergence Comparison: Model 1 vs Model 3

Diagnostic	**Model 1**	**Model 3**	**Status**
Multivariate PSRF	1.04	1.01	✓
Max individual $\hat{R}$	1.06	1.01	✓
Min ESS (coefficients)	110.0	79.5	Model 1
Min ESS (batch effects)	46.2	106.1	Model 3 ✓
ESS (dispersion r)	13168.3	60852.6	Model 3 ✓
ESS (batch variance)	14498.9	132073.2	Model 3 ✓
Iterations	30000	300000	Model 3

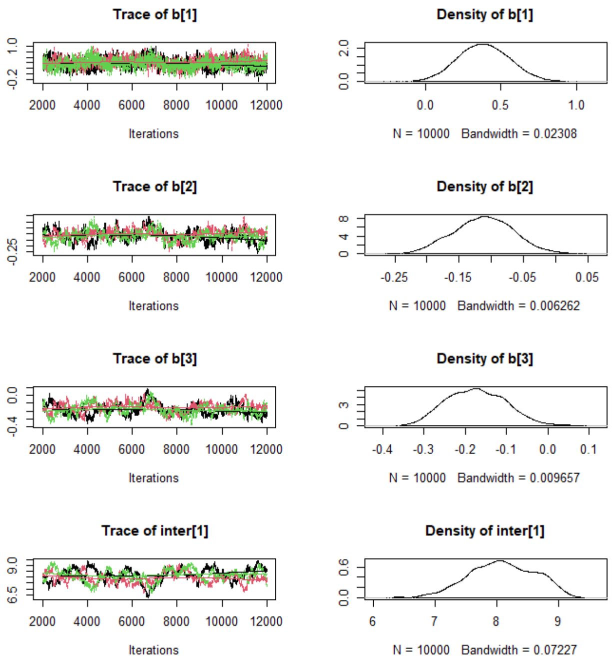


Figure 4: Example of trace plots for model 1

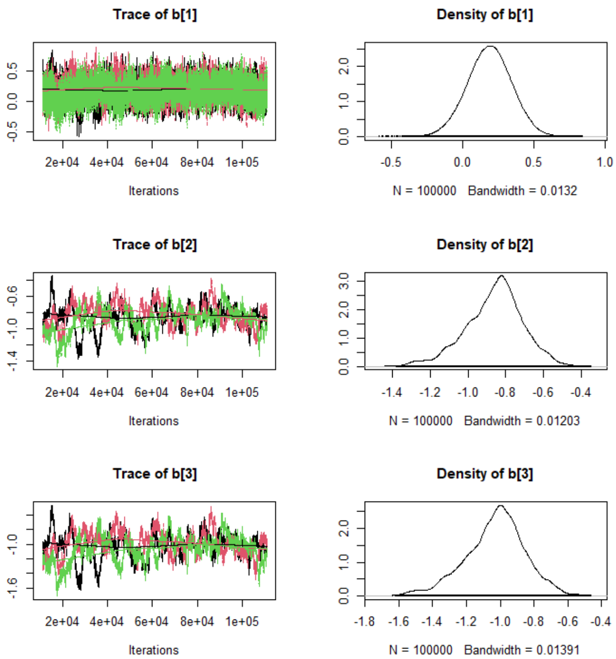


Figure 5: Example of trace plots for model 3

The higher iterations (300,000 vs. 30,000) in Model 3 contribute to these improvements, but the dramatic gains in effective sample size indicate that Model 3's structure also enhances MCMC mixing. ***Note that the trace plots indicate high autocorrelation in both models and specially in model 3. This is generally problematic, but because our aim is not to provide confidence interval but to predict the the influence of the charging policy on the number of life cycles we accept this and use this models.***

3.2 Observations

From the summary of the models we can see that the charge cut-off contribute with a positive coefficient while the charging rate and discharging rates contribute with a negative coefficients. The mixing term in the model 3 contribute positively. This indicates that charging for higher cut-off increase life cycle while while higher charging and discharging rates reduce it.

3.3 Model Comparison

In this section we compare the performance of the three models.

Table 12: Deviance Information Criterion

	Model 1	Model 2	Model 3
DIC	1133	1104	1105

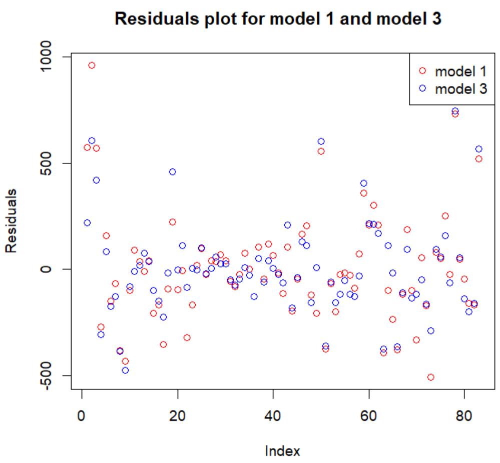
Table 12, summaries the deviance information criterion (DIC) for each of the three models. It is clear that models 2 and 3 out preform model 1.

3.4 Residuals

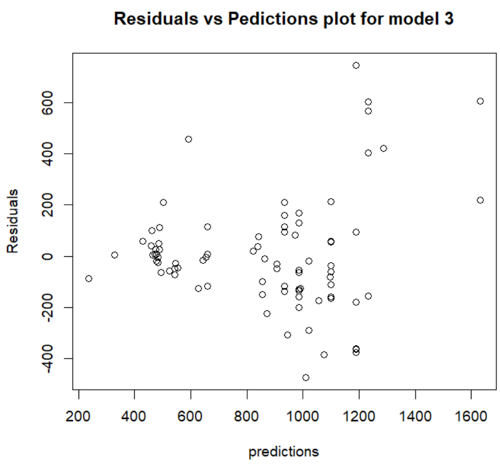
In this section we analyze the residuals for model 1 and 3 for the fitting data. 6a shows the residuals for the models. The residuals are centered at zero, however we observe the presence of large residuals. 6c, shows that the variance of the residuals increase as the predicted value increase, while 6b shows the same behavior for model 3 but on a smaller scale. Figures 7a and 7b depict the residuals per batch for model 1 and 3 respectively. They show that model 3 residues the values 0 the residuals in accordance with the values of the DIC. Finally we test model 3 using the testing data set that was not used in fitting. The residuals are reported in 8b and 8a, we see that the behavior of residuals is consistent between the fitting and testing data which indicates that model 3 can be generalized for different data sets.

4 Prior Sensitivity Analysis

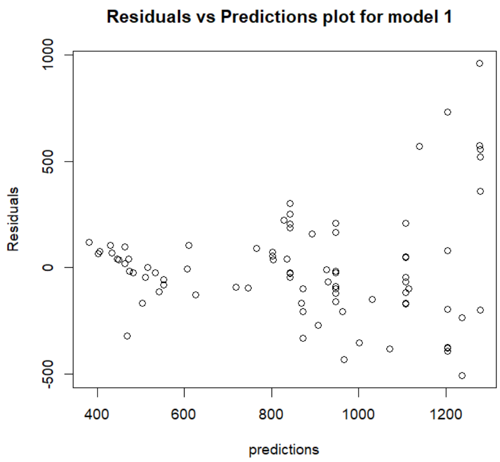
In this sections we analyse the influence of the priors on the estimated values of the models parameters mainly the β parameters. Figure 9 shows the priors used in models 2 and 3 for the β parameters. Models 2 have skeptical priors with most of the probability centered at 0, while model 3 have a more spread out probability. The posterior probability densities for each of the parameters are reported in figure 10. We can see that the densities are closes for the two different types of priors strengthening our believe in the validity of the results.



(a) Residuals: Fitting Data

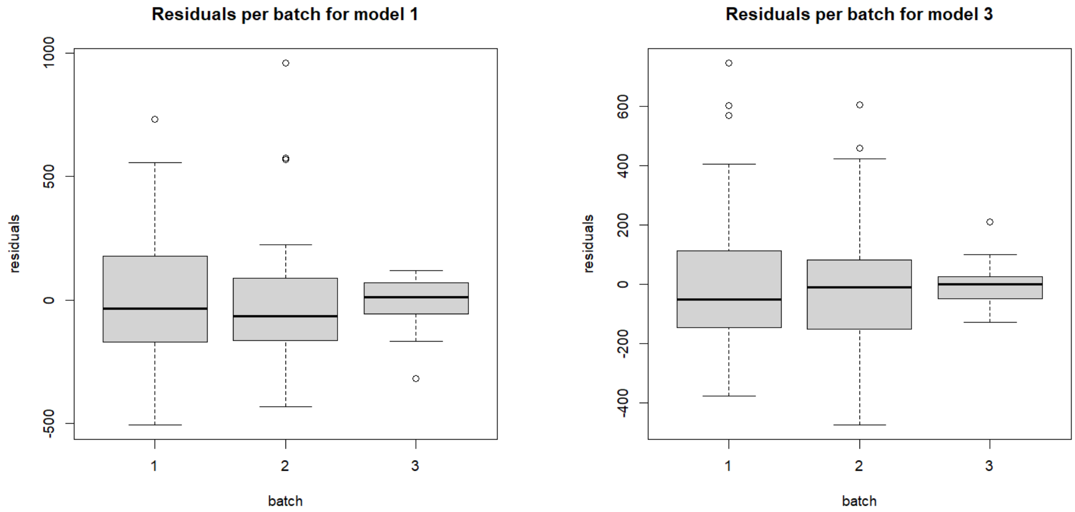


(b) Residuals vs Predictions (Model 3)



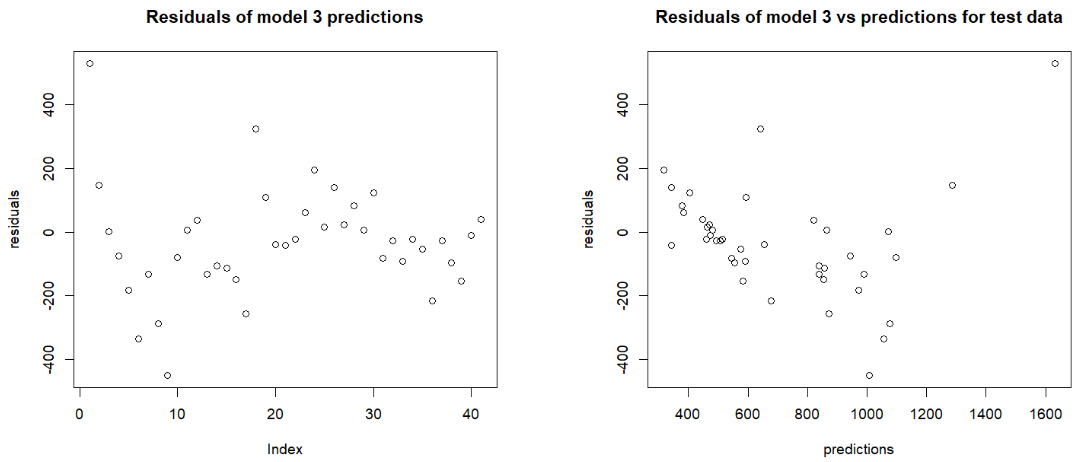
(c) Residuals vs Predictions (Model 1)

Figure 6: Model Diagnostic: Residual Analysis



(a) Residuals per batch for model 1 (b) Residuals per batch for model 3

Figure 7: Box plots for residuals for each batch



(a) Residuals (b) Residuals against predictions

Figure 8: Residuals of model 3 for testing data

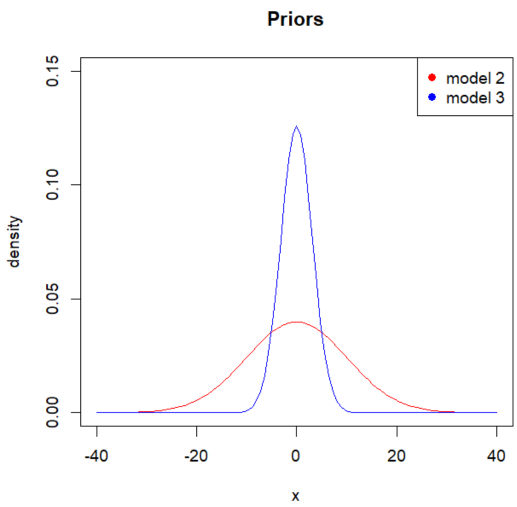


Figure 9: Priors for model 2 and model 3

5 Model 3 predictions

Given the results from the two previous section we propose to use model 3 to answer the analysis questions. What is the probability that a battery under a given charging policy achieve a given targets of life cycles. The policy we considered in the analysis are reported in table 1. We sample for each MCMC result thus we take into account uncertainty in predictions.

Figure 11 shows the probability of attaining a target life cycles for each of the three charging policy. We can see that for the aggressive charging policy a the probability for a battery to get to 1000 life cycles is around 40%, while it is about 55% for a balanced charging policy and around 70% for the conservative policy.

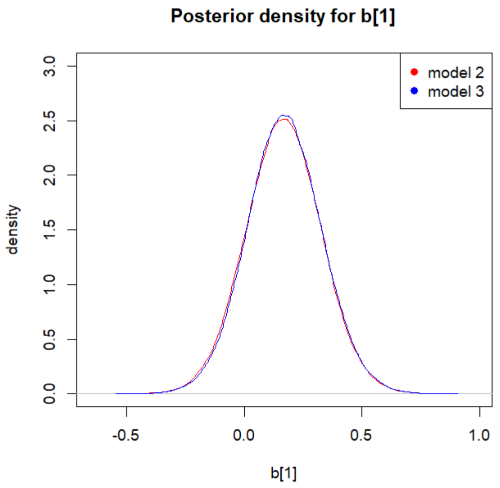
6 Conclusion

This analysis developed a Bayesian hierarchical negative binomial regression model to quantify the influence of charging policy on lithium-ion battery cycle life, with explicit uncertainty quantification for risk-aware decision-making in energy storage applications.

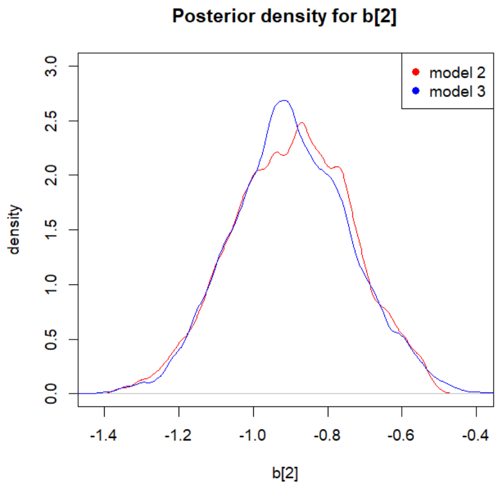
6.1 Summary of Findings

Our data exploration identified strong overdispersion in cycle-life counts (variance-to-mean ratios ranging from 12–174 across batches), justifying the negative binomial likelihood over Poisson regression. Significant batch-level variability in median cycle life (1032 and 921 cycles for batches 1 and 2 versus 473 for batch 3) motivated a hierarchical random-intercept structure.

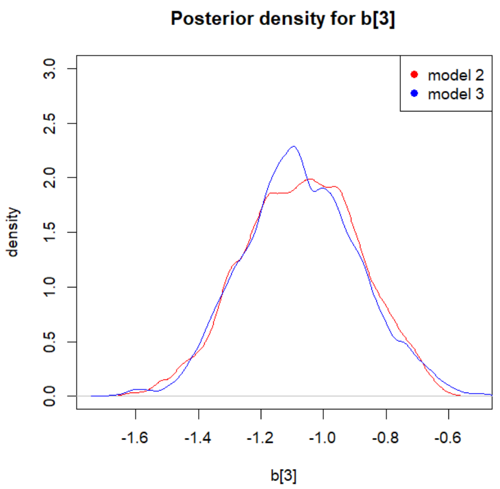
Model 3, incorporating an interaction term between charging and discharging rates with 300,000 MCMC iterations, demonstrated superior convergence (multivariate $\hat{R} = 1.01$) and effective sample sizes (ESS for batch variance = 132,073). The posterior estimates reveal:



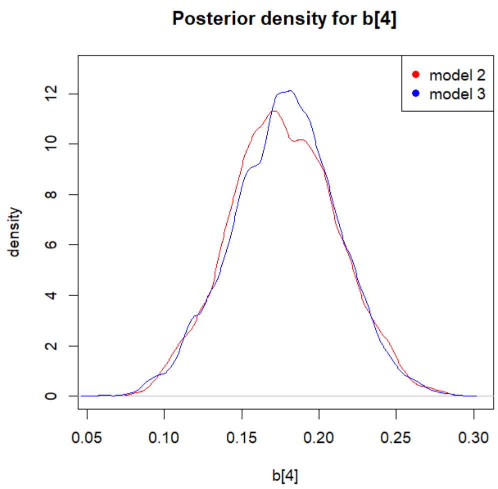
(a) Posterior density for β_1



(b) Posterior density for β_2



(c) Posterior density for β_3



(d) Posterior density for β_4

Figure 10: Posterior probability density for β

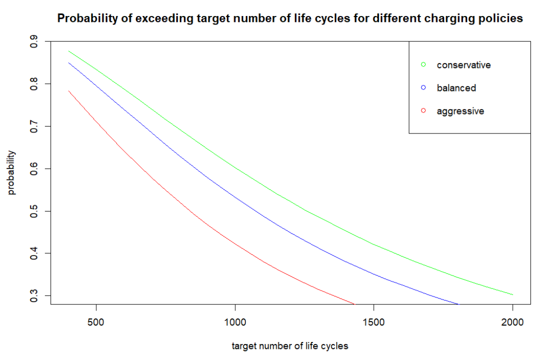
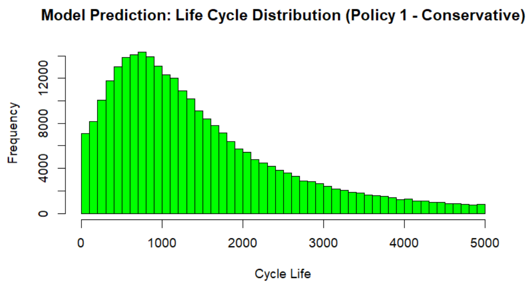
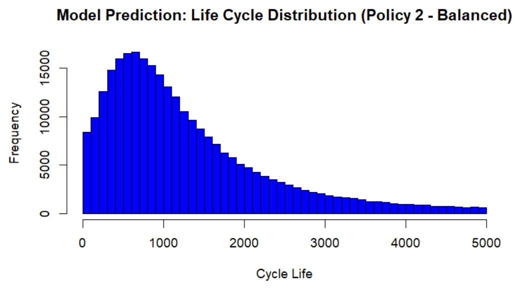


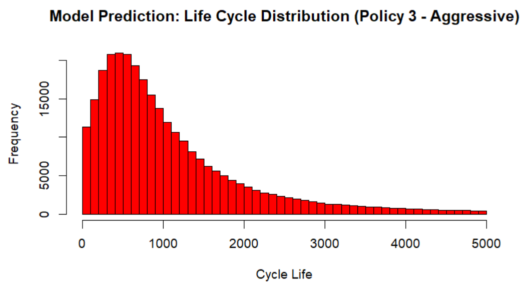
Figure 11: Probability of attaining a target number of life cycles



(a) Histogram for conservative policy



(b) Histogram for balanced policy



(c) Histogram for aggressive policy

Figure 12: Probability histograms for the three different policies

- **Charging cut-off** ($\beta_1 = 0.189$, 95% CI: $[-0.122, 0.492]$): Positive effect; higher depth-of-discharge increases cycle life
- **Charging rate** ($\beta_2 = -0.861$, 95% CI: $[-1.199, -0.578]$): Strong negative effect; faster charging reduces cycle life
- **Discharging rate** ($\beta_3 = -1.029$, 95% CI: $[-1.419, -1.015]$): Strong negative effect; faster discharging significantly reduces cycle life
- **Charge-discharge interaction** ($\beta_4 = 0.171$, 95% CI: $[0.109, 0.241]$): Positive interaction; moderate discharge rates can partially mitigate fast charging stress

Model validation on held-out test data confirmed generalization: residual patterns remained consistent between training and test sets, supporting deployment confidence.

6.2 Policy Comparison and Recommendations

Table 13 synthesizes the performance of the three charging policies under operationally relevant cycle-life targets.

6.2.1 Key Insights

1. **Probability gap widens at higher targets:** At 1000-cycle threshold, the probability gap between conservative and aggressive policies is 30 percentage points. This gap expands to 20

Table 13: Comprehensive Policy Comparison: Probability of Achieving Target Cycle Life

Metric	Conservative	Balanced	Aggressive	Tradeoff
Charging Parameters				
Charging rate	3.6C	4.4C	5.4C	Faster = shorter life
Charging cut-off (% DoD)	40%	60%	80%	Deeper = longer life
Discharging rate	3.6C	3.6C	4.8C	Faster = shorter life
Probabilistic Predictions (Model 3)				
P(Cycle life ≥ 500)	98%	96%	88%	Δ = 10%
P(Cycle life ≥ 800)	90%	81%	65%	Δ = 25%
P(Cycle life ≥ 1000)	70%	55%	40%	Δ = 30%
P(Cycle life ≥ 1500)	25%	12%	5%	Δ = 20%
Operational Characteristics				
Thermal stress	Low	Moderate	High	Impacts cooling requirements
Charging time per cycle	High	Moderate	Low	Throughput vs. longevity
Grid flexibility	Limited	Good	High	Can respond to demand peaks

percentage points at 1500 cycles, highlighting the compounding effect of thermal stress over extended operation.

2. **Charging rate dominates discharging rate:** Both charging and discharging rates show negative coefficients, but the charging-rate coefficient ($\beta_2 = -0.861$) is comparable in magnitude to the discharging coefficient ($\beta_3 = -1.029$). Fast charging is nearly as harmful as fast discharging, suggesting thermal management during charging is critical.

3. **The interaction term enables tactical optimization:** The positive interaction ($\beta_4 = 0.171$) suggests that faster discharging can partially compensate for moderate charging stress. A balanced policy charging at 4.4C with 3.6C discharge achieves 55% probability at 1000 cycles — intermediate between aggressive and conservative, offering a practical compromise.

4. **Batch variability is substantial:** Posterior estimates of batch intercepts range from $\alpha_3 = 11.29$ to $\alpha_1 = 11.96$, with posterior standard deviation $\sigma_\alpha = 0.90$. This $\approx$ 8% variation in baseline cycle life reflects manufacturing process differences and motivates quality control in cell fabrication.

6.3 Implications for Energy Storage Operators

The probabilistic framework enables risk-quantified policy selection:

- **Baseload storage systems** (daily or weekly cycling) should adopt conservative or balanced policies to maximize equipment lifespan and minimize replacement cycles.
- **Peak shaving applications** (high-frequency cycling) can justify aggressive policies if the value of fast charge/discharge response exceeds the cost of accelerated degradation.
- **Hybrid strategies** — e.g., conservative cycling during low-demand periods and aggressive cycling during peak hours — may optimize total value delivery.

The quantified uncertainty (e.g., 70% vs. 65% confidence intervals) allows grid operators to embed these predictions into reserve margin calculations and maintenance scheduling without over-committing or over-provisioning.

6.4 Limitations and Future Work

Limitations:

1. Data limited to LFP/graphite chemistry; NCA and NMC chemistries may exhibit different degradation mechanisms and require separate models.

2. Model does not account for temperature-dependent cycling; real-world performance depends on thermal management. A hierarchical model incorporating ambient temperature and active cooling would strengthen predictions.

3. Autocorrelation in posterior samples (visible in trace plots) suggests potential for algorithmic improvement (e.g., tempering, adaptive Metropolis-Hastings).

4. Predictions assume stochastic cycling at fixed policy parameters; time-varying or adaptive charging profiles are not modeled.

Future directions:

1. **Multi-chemistry comparison:** Fit the same hierarchical framework to NCA and NMC datasets to quantify chemistry-specific degradation rates.

2. **Thermal dynamics coupling:** Integrate heat generation models (Joule heating, kinetic heating during intercalation) with the statistical framework to predict joint cycle-life and temperature evolution.

3. **Real-time monitoring:** Deploy the fitted model as a Bayesian filter to estimate remaining useful life from early-cycle capacity data, updating predictions as new information arrives.

4. **Optimization under uncertainty:** Formulate a stochastic control problem to find the optimal dynamic charging policy that maximizes expected grid revenue while maintaining acceptable cycle-life confidence levels.

6.5 Conclusion

This analysis demonstrates that Bayesian hierarchical modeling is a principled approach to quantifying battery cycle-life uncertainty under operational charging policies. The framework successfully translates data variability into actionable probabilistic predictions, enabling grid operators and battery manufacturers to make risk-aware decisions. Model 3 achieved superior convergence and generalization, providing a robust foundation for deployment in energy storage optimization workflows.

The empirical finding that conservative policies achieve 70% probability of $\geq$ 1000 cycles versus 40% for aggressive policies quantifies a well-known engineering tradeoff; the Bayesian posterior distribution framework enables cost-benefit analysis that was previously opaque. Future work incorporating chemistry variation, thermal dynamics, and real-time state estimation will extend the applicability to heterogeneous grid deployments.

References

References

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